KON429E – Autonomous Vehicles
Homework 1 Report

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Abstract

This report presents the modeling, implementation, and analysis of a nonlinear bicycle vehicle model in Simulink for the first homework of the KON429E Autonomous Vehicles course. The model includes kinematics, longitudinal dynamics, lateral dynamics, longitudinal slip calculation, tire–chassis projection, and a linear tire model with force saturation. In Part (a), the model is validated using the given steering and torque profiles, and the obtained trajectory, position coordinates, yaw angle, and velocity responses are compared with the reference outputs provided in the homework statement. In Part (b), a more aggressive steering and torque input is applied in order to generate a spin-like unstable vehicle response. The resulting trajectory, yaw response, velocity, tire slip quantities, and tire forces are analyzed and interpreted in detail.

1. Introduction

Vehicle motion can be represented at different levels of complexity depending on the intended application. For control-oriented studies, the bicycle model is one of the most widely used reduced-order representations, since it captures the essential longitudinal and lateral vehicle behavior while remaining computationally efficient. In this homework, a nonlinear bicycle model was implemented in Simulink. The model inputs are the front steering angle δ_f and the torque applied to the front axle T_f, whereas the outputs are the global vehicle coordinates X, Y, and the yaw angle ψ.

The homework required two main tasks. First, a vehicle model had to be built and validated using the provided parameters and the supplied reference input profile. Second, a new input configuration had to be designed so that the vehicle exhibits drifting or spinning behavior. In this report, both tasks are addressed systematically.

2. Problem Statement and Objectives

The main objectives of the homework are listed below:

1) Build a nonlinear bicycle vehicle model in Simulink using the symbolic structure provided in the lecture material.

2) Use a linear tire model with saturation for both longitudinal and lateral tire forces.

3) Validate the model using the given sample steering and torque inputs.

4) Obtain the trajectory, $X(t)$, $Y(t)$, $\psi(t)$, and $V(t)$ responses and compare them with the reference plots.

5) Find a different steering and torque configuration that results in a drifting or spinning response.

6) Analyze the resulting instability using slip angles, slip ratios, and tire force behavior.

3. Vehicle Model Formulation

3.1. State Variables

The implemented vehicle model is based on the following dynamic states:

$$\mathbf{x} = \begin{bmatrix} V & \beta & r & X & Y & \psi & \omega_f & \omega_r \end{bmatrix}^T \tag{1}$$

where:

- V is the vehicle speed,
- β is the vehicle body slip angle,
- $r = \dot{\psi}$ is the yaw rate,
- X and Y are the global position coordinates,
- ψ is the yaw angle,
- ω_f and ω_r are the equivalent front and rear wheel angular speeds.

3.2. Trajectory Kinematics

The global motion of the vehicle is calculated from the vehicle speed and orientation as

$$\dot{X} = V\cos(\psi + \beta), \tag{2}$$
$$\dot{Y} = V\sin(\psi + \beta), \tag{3}$$
$$\dot{\psi} = r. \tag{4}$$

These equations map the body-fixed motion of the vehicle to the inertial frame.

3.3. Longitudinal and Lateral Dynamics

Using the nonlinear formulation presented in the lecture notes, the vehicle dynamics are written as

$$\dot{V} = \frac{F_x \cos\beta + F_y \sin\beta}{m}, \tag{5}$$

$$\dot{\beta} = \frac{-F_x \sin\beta + F_y \cos\beta}{mV} - r, \tag{6}$$

$$\dot{r} = \frac{M_z}{J}. \tag{7}$$

Here, m is the vehicle mass and J is the yaw moment of inertia. The total longitudinal force F_x, lateral force F_y, and yaw moment M_z are obtained by combining the tire forces in the chassis frame.

3.4. Slip Angle Calculation

The body-fixed longitudinal and lateral velocity components are defined as

$$u = V\cos\beta, \tag{8}$$

$$v = V\sin\beta. \tag{9}$$

Using these quantities, the front and rear slip angles are calculated directly as

$$\alpha_f = \tan^{-1}\left(\frac{v + rl_f}{u}\right) - \delta_f, \tag{10}$$

$$\alpha_r = \tan^{-1}\left(\frac{v - rl_r}{u}\right) - \delta_r, \tag{11}$$

where l_f and l_r are the distances from the center of gravity to the front and rear axles, respectively.

3.5. Longitudinal Slip Ratio Calculation

The wheel-frame longitudinal contact speeds are computed as

$$V_{fx} = u\cos\delta_f + (v + rl_f)\sin\delta_f, \tag{12}$$

$$V_{rx} = u\cos\delta_r + (v - rl_r)\sin\delta_r. \tag{13}$$

The longitudinal slip ratios are then calculated piecewise depending on whether the wheel is in driving or braking mode:

$$S_{Lf} = \frac{\omega_f R - V_{fx}}{\max(\omega_f R, \epsilon)} \quad \text{(driving)} \tag{14}$$

or

$$S_{Lf} = \frac{\omega_f R - V_{fx}}{\max(V_{fx}, \epsilon)} \quad \text{(braking)} \tag{15}$$

and similarly for the rear wheel. Here, R is the effective wheel radius and ϵ is a small number used to avoid division by zero.

3.6. Linear Tire Model with Saturation

The homework required the use of a linear tire model with saturation. Therefore, the tire forces are computed as

$$F_{xf} = \text{sat}(C_{xf} S_{Lf}), \tag{16}$$
$$F_{xr} = \text{sat}(C_{xr} S_{Lr}), \tag{17}$$
$$F_{yf} = \text{sat}(C_{yf} \alpha_f), \tag{18}$$
$$F_{yr} = \text{sat}(C_{yr} \alpha_r), \tag{19}$$

where $C_{xf}, C_{xr}, C_{yf}, C_{yr}$ are the longitudinal and lateral stiffness values, and the saturation limits are determined by the maximum longitudinal and lateral tire forces.

During debugging and validation, it was found that the axle total forces must be represented by doubling the single-wheel tire force contributions in the implemented model. This scaling significantly improved the match with the reference outputs.

3.7. Tire–Chassis Projection

The tire forces are projected into the vehicle body frame as

$$F_x = -\sin\delta_f F_{yf} - \sin\delta_r F_{yr} + \cos\delta_f F_{xf} + \cos\delta_r F_{xr} - F_{load}, \tag{20}$$
$$F_y = \cos\delta_f F_{yf} + \cos\delta_r F_{yr} + \sin\delta_f F_{xf} + \sin\delta_r F_{xr}, \tag{21}$$
$$M_z = l_f \cos\delta_f F_{yf} - l_r \cos\delta_r F_{yr} + l_f \sin\delta_f F_{xf} - l_r \sin\delta_r F_{xr}. \tag{22}$$

Since the given rear steering input is zero, $\delta_r = 0$ in all simulations.

3.8. Resistive Forces

The total resistive load is composed of aerodynamic drag, rolling resistance, and road grade effect:

$$F_{load} = F_{aero} + F_{roll} + F_{grade}. \tag{23}$$

The aerodynamic drag is modeled as

$$F_{aero} = \frac{1}{2}\rho C_d A\, V|V|, \tag{24}$$

which ensures that drag always opposes the direction of motion. The rolling resistance is modeled in a direction-consistent way as

$$F_{roll} = C_{rr} mg \tanh\left(\frac{V}{V_\varepsilon}\right), \tag{25}$$

so that it does not artificially pull the vehicle backward at standstill. The grade resistance term is

$$F_{grade} = mg \sin(\theta), \tag{26}$$

where θ is the road slope. In this homework, the road is flat, thus $\theta = 0$.

4. Vehicle Parameters

The parameters supplied in the homework file are summarized in Table 1.

Table 1: Vehicle parameters used in the Simulink model.

Parameter	Symbol	Value
Front axle distance	l_f	1.4 m
Rear axle distance	l_r	1.1 m
Vehicle mass	m	1300 kg
Yaw inertia	J	580 kg m^2
Wheel inertia	J_t	3.5 kg m^2
Wheel radius	R	0.3 m
Air density	ρ	1.225 kg/m^3
Drag coefficient	C_d	0.3
Frontal area	A	2.0 m^2
Rolling resistance coefficient	C_{rr}	0.014
Front longitudinal stiffness	C_{xf}	55 000 N
Rear longitudinal stiffness	C_{xr}	55 000 N
Front lateral stiffness	C_{yf}	45 000 N/rad
Rear lateral stiffness	C_{yr}	45 000 N/rad
Maximum longitudinal tire force	$F_{max,x}$	15 000 N
Maximum lateral tire force	$F_{max,y}$	9000 N

5. Simulink Implementation

The Simulink model was constructed in modular form using separate subsystems for:

- kinematics and geometry,
- longitudinal slip calculation,
- tire model,
- tire–chassis projection,

- resistive force calculation,
- vehicle dynamics,
- trajectory calculation,
- wheel rotational dynamics.

The solver settings recommended in the homework statement were used. The simulation was run with a fixed-step solver and a step size of 0.001 s.

6. Part (a): Model Validation with the Given Input

6.1. Reference Input

The supplied reference input consists of a front steering command $\delta_f(t)$ and a front axle torque input $T_f(t)$. The rear steering and rear torque inputs are zero. This input profile was used to validate the Simulink model.

6.2. Validation Procedure

The model was iteratively debugged until the main responses matched the given reference figures with good accuracy. The most important corrections during this process were:

- correcting the sign-consistent resistive force model,
- using direct wheel-frame longitudinal speed components for slip ratio computation,
- using direct slip-angle computation based on body-fixed velocity components,
- scaling the tire forces to represent axle total forces.

After these corrections, the final outputs matched the reference trajectories and response shapes very closely.

6.3. Results for Part (a)

The validated simulation results are shown in Figures 1–5.

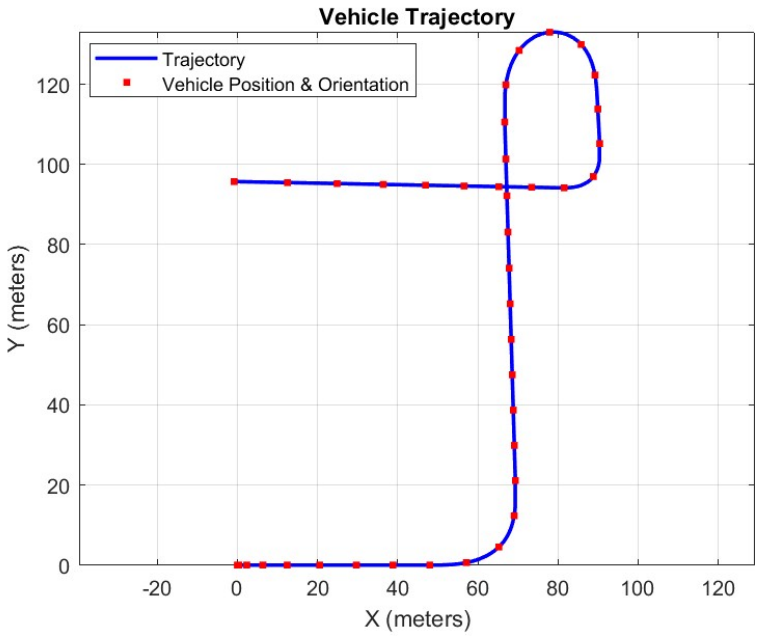


Figure 1: Vehicle trajectory obtained for the reference input. The result closely matches the target trajectory provided in the homework statement.

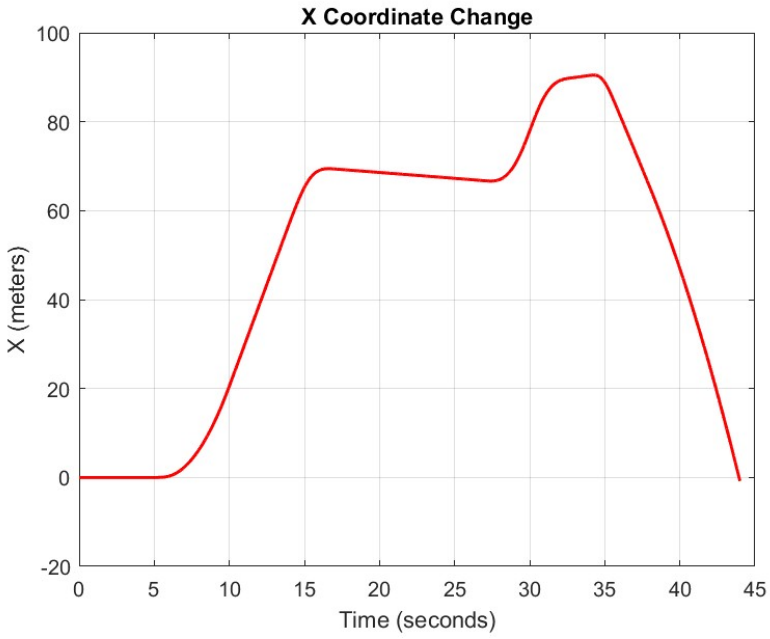


Figure 2: Global X coordinate response for the reference input.

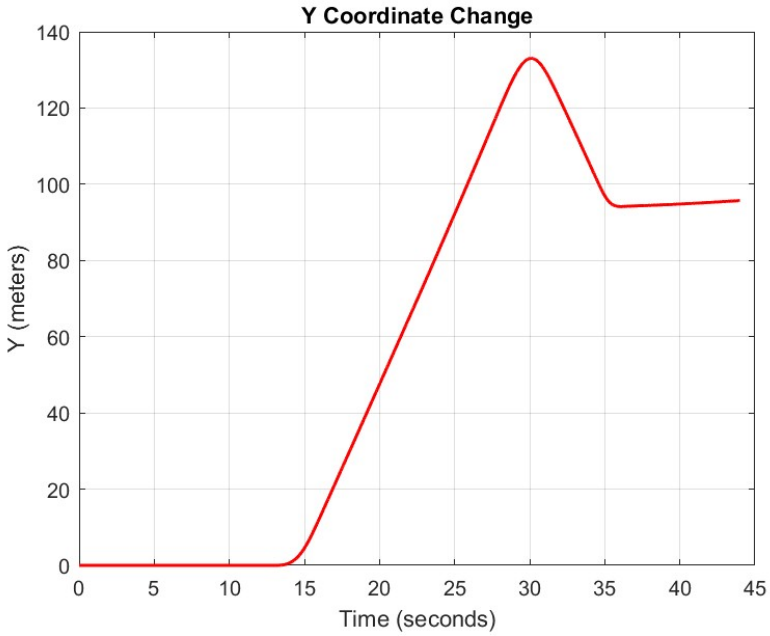


Figure 3: Global Y coordinate response for the reference input.

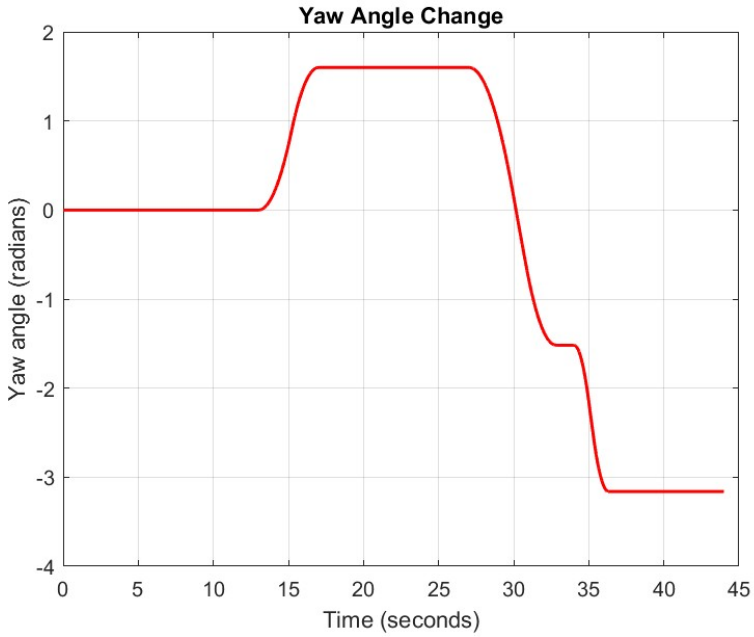


Figure 4: Yaw angle response for the reference input.

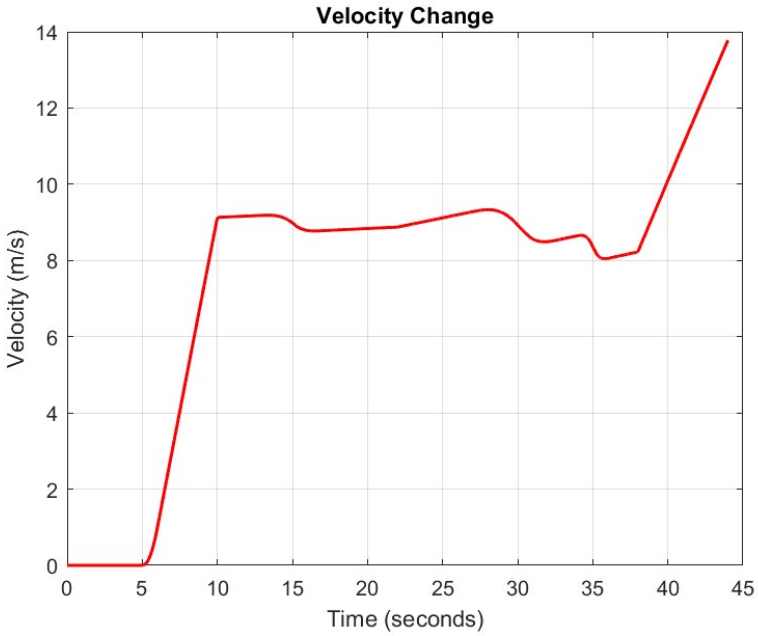


Figure 5: Vehicle speed response for the reference input.

The final numerical values obtained from the validated model are listed below:

$$X_{final} \approx -0.82\,\text{m},$$
$$Y_{final} \approx 95.72\,\text{m},$$
$$\psi_{final} \approx -3.16\,\text{rad},$$
$$V_{final} \approx 13.78\,\text{m/s}.$$

These values are in close agreement with the reference plots given in the homework handout.

6.4. Discussion of Part (a)

The validated model reproduces the expected vehicle path, yaw development, and speed evolution successfully. The trajectory includes the initial straight section, the nearly vertical motion segment, the upper turning loop, and the final return toward the left side, consistent with the target behavior. This confirms that the nonlinear bicycle model, together with the tire and slip formulations, is able to capture the required vehicle response.

7. Part (b): Drifting / Spinning Case

7.1. Objective

The second part of the homework requires an input configuration that produces drifting or spinning behavior. In this study, several candidate steering and torque profiles were tested. Highly aggressive inputs were found to drive the system into strong instability, including very large slip angles and severe yaw excursions.

The final selected case is best described as a **spin-like unstable response** rather than a smooth controlled drift. The selected input produced a rapid growth of yaw angle, loss of normal path tracking, and a substantial deviation from the nominal motion pattern.

7.2. Chosen Input Profile

For this part, a more aggressive steering and propulsion profile than the reference case was applied. The vehicle was first accelerated to a relatively high speed, and then a sharp steering maneuver was introduced. This resulted in a large increase in front and rear slip angles, causing the lateral tire forces to approach their limits and leading to spin-like behavior.

7.3. Results for Part (b)

The corresponding outputs are shown in Figures 6–9.

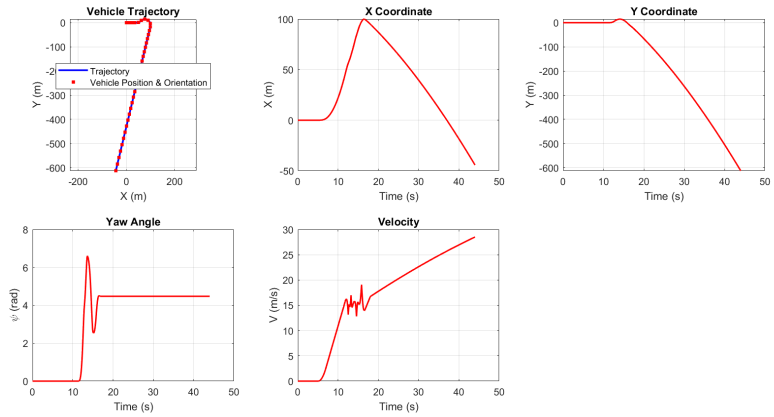


Figure 6: Main outputs for the aggressive input case: trajectory, $X(t)$, $Y(t)$, yaw angle, and velocity. The trajectory deviates significantly from the nominal reference path, indicating loss of lateral stability.

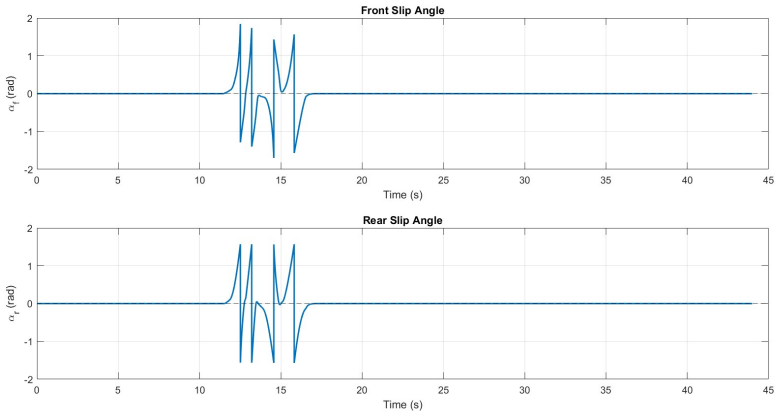


Figure 7: Front and rear slip angles for the aggressive case. Large temporary peaks indicate severe tire slip and instability onset.

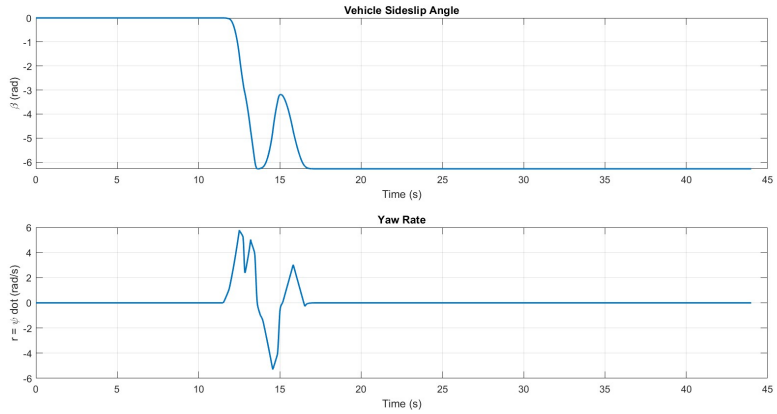


Figure 8: Vehicle body slip angle and yaw rate responses for the aggressive case. The rapid yaw-rate growth confirms the spin-like unstable response.

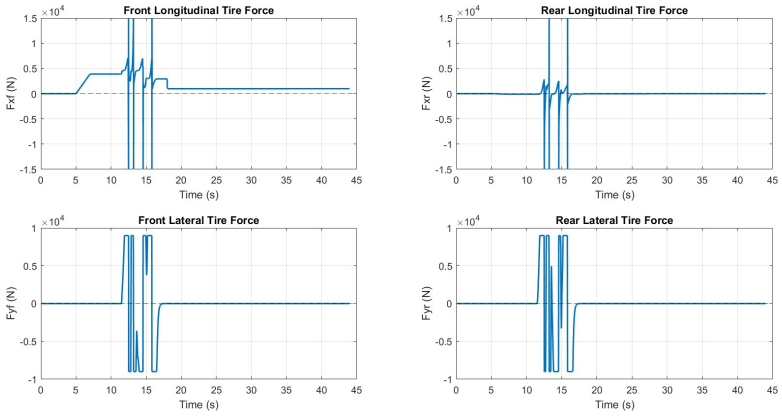


Figure 9: Tire force responses for the aggressive input case. The tire forces approach their limits during the instability phase.

7.4. Interpretation of the Spin-Like Response

The aggressive steering command, applied at elevated vehicle speed, causes the front and rear slip angles to increase rapidly. As the slip angles grow, the lateral tire forces approach their saturation limits. Under these conditions, the vehicle loses its normal lateral stability and experiences a rapid yaw excursion.

The yaw angle plot shows a sudden and strong rotation, which indicates that the vehicle is no longer following the nominal path. Similarly, the trajectory plot shows that the vehicle departs significantly from the original motion pattern and continues along a new direction after the instability. This is characteristic of a spin-like response.

From the tire force and slip-angle plots, it can be concluded that the instability is caused by excessive lateral demand relative to the available tire force. In other words, the steering input is too aggressive for the speed of the vehicle, and the tire model predicts the onset of large slip and force saturation. Therefore, the observed response can be interpreted as a spinning or severe loss-of-stability event.

8. General Discussion

This homework demonstrates that even a reduced-order nonlinear bicycle model can reproduce rich vehicle dynamics when tire slip and force saturation are included. In Part (a), the model successfully tracks the nominal reference motion when the correct tire scaling and resistive force formulation are used. In Part (b), the same model also reveals how aggressive steering at high speed can drive the vehicle into instability.

A particularly important modeling lesson from this homework is that sign consistency and physical interpretation of the force terms are critical. During model development, the initial mismatch with the reference solution was traced primarily to the resistive force formulation and the effective scaling of the tire forces. Once these were corrected, the model outputs aligned closely with the reference plots.

Another important observation is that a simple linear tire model with saturation can reproduce the onset of instability, but under extremely aggressive inputs it may also produce abrupt switching-type responses. Therefore, the model is suitable for demonstrating stability loss and spin-like behavior, but a smoother nonlinear tire model would be preferable for more detailed high-slip drift analysis.

9. Conclusion

In this study, a nonlinear Simulink bicycle model was developed for the KON429E Autonomous Vehicles homework. The model included kinematics, longitudinal and lateral dynamics, slip-angle and slip-ratio calculations, resistive forces, tire force generation, and tire–chassis projection.

For Part (a), the model was validated successfully against the supplied reference input. The obtained trajectory, position coordinates, yaw angle, and vehicle speed were found to be in close agreement with the target outputs provided in the homework statement.

For Part (b), an aggressive steering and torque configuration was tested in order to provoke instability. The resulting response exhibited a spin-like behavior characterized by large slip angles, rapid yaw motion, and a strong deviation from the nominal path. This behavior was analyzed using the trajectory, yaw response, vehicle speed, slip-angle plots, and tire force plots.

Overall, the homework provided a useful practical demonstration of how nonlinear vehicle dynamics, tire slip, and steering inputs interact to determine vehicle stability and maneuvering behavior.

Appendix: Chosen Input Profile

```
%% Time Vector
time = (0:0.001:24)';

%% Steering Angle Profile
delta_f_val = zeros(size(time));

for i = 1:length(time)
    t = time(i);

    if t < 11.5
        delta_f_val(i) = 0;

    elseif t < 12.2
        % sola hizli giris
        delta_f_val(i) = 0.28 * ((t - 11.5)/0.7);

    elseif t < 13.0
        % kisa sure tut
        delta_f_val(i) = 0.28;

    elseif t < 13.8
        % hizli counter steer
        delta_f_val(i) = 0.28 - 0.43*((t - 13.0)/0.8);

    elseif t < 14.5
        % kisa sag tut
        delta_f_val(i) = -0.15;

    elseif t < 15.2
        % merkeze don
        delta_f_val(i) = -0.15 + 0.15*((t - 14.5)/0.7);

    else
        delta_f_val(i) = 0;
    end
end

delta_f = [time, delta_f_val];
delta_r = [time, zeros(size(time))];

%% Torque Profile
torque_val = zeros(size(time));

```

```
for i = 1:length(time)
    t = time(i);

    if t < 5
        torque_val(i) = 0;

    elseif t < 7
        torque_val(i) = 1200 * ((t - 5)/2);

    elseif t < 11.5
        torque_val(i) = 1200;

    elseif t < 14.5
        torque_val(i) = 1400;

    elseif t < 18
        torque_val(i) = 900;

    else
        torque_val(i) = 300;
    end
end

Tf = [time, torque_val];
Tr = [time, zeros(size(time))];
```

Listing 1: MATLAB script used for Part (b)